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|---|---|
| Experiment No. | 5 |
| Experiment Title | Breast Cancer Diagnosis using Decision Tree and Random Forest Classifiers |
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# 1 Objective

The objective of this experiment is to build and compare two tree-based supervised classification algorithms, a Decision Tree and a Random Forest, for diagnosing breast tumours as malignant or benign from digitized cell-nuclei measurements. The experiment covers exploratory data analysis, hyperparameter tuning through grid search, 5-fold cross-validation, evaluation using standard classification metrics and ROC-AUC, and an analysis of how tree depth affects overfitting.

# 2 Problem Statement

Given 30 real-valued features computed from a digitized image of a fine needle aspirate (FNA) of a breast mass (mean, standard error and worst value of nucleus radius, texture, perimeter, area, smoothness, etc.), the task is to predict whether the tumour is malignant (class label 1) or benign (class label 0). This is a binary classification problem. The input is a 30-dimensional feature vector per sample, and the output is a binary diagnosis label. The goal is to determine which of the two tree-based classifiers gives the more reliable diagnosis on unseen data.

# 3 Dataset Description

| | |
|---|---|
| Dataset Name | Wisconsin Diagnostic Breast Cancer (WDBC) |
| Dataset Source | UCI Machine Learning Repository (loaded locally as `wdbc.data`) |
| Number of Samples | 569 |
| Number of Features | 30 numerical features (the `id` column was dropped as it is not predictive) |
| Number of Classes | 2 (0 = Benign, 1 = Malignant) |
| Missing Values | None (the WDBC feature set is complete; no imputation was performed) |
| Train-Test Split | 80% train / 20% test (455 training samples, 114 test samples), `random_state=42` |

# 4 Exploratory Data Analysis

## Dataset Overview

After loading `wdbc.data` with manually assigned column names and dropping the non-predictive `id` column, the dataset contains 569 rows and 31 columns (30 features plus the `diagnosis` target). The `diagnosis` column, originally categorical (`B`/`M`), was label-encoded so that Benign = 0 and Malignant = 1.

## Statistical Summary

The first six features (mean radius, texture, perimeter, area, smoothness and compactness) show strong positive correlation with the diagnosis label, particularly radius, perimeter and area, which are geometrically related and therefore highly correlated with one another. This is consistent with the biological expectation that larger, more irregular cell nuclei are associated with malignancy.

## Missing Value Analysis

The WDBC dataset is a clean, complete dataset with no missing entries in any of the 30 features, so no missing-value handling step was required before model training.

## Class Distribution

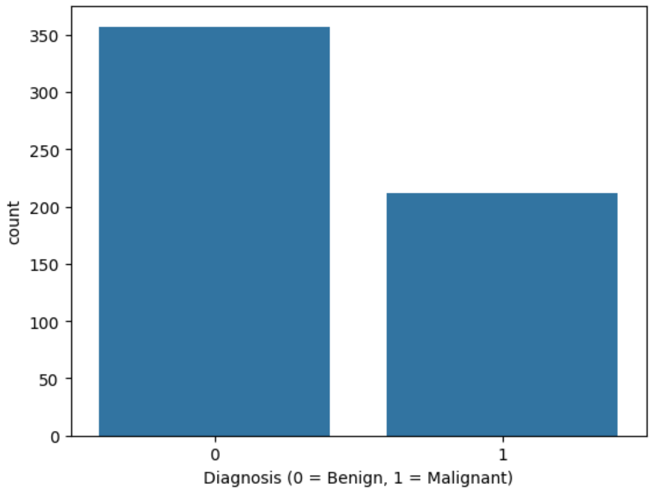


Figure 1: Class distribution of the WDBC dataset (0 = Benign, 1 = Malignant).

**Inference:** The dataset contains 357 benign and 212 malignant cases, showing a moderate class imbalance in favour of benign tumours. This imbalance is not severe enough to require resampling, but it does mean that precision and recall must be examined alongside accuracy to properly judge how well each model identifies malignant cases. The imbalance also motivates the use of stratifiable evaluation techniques such as cross-validation, which was applied in later stages of this experiment.

### Correlation Matrix

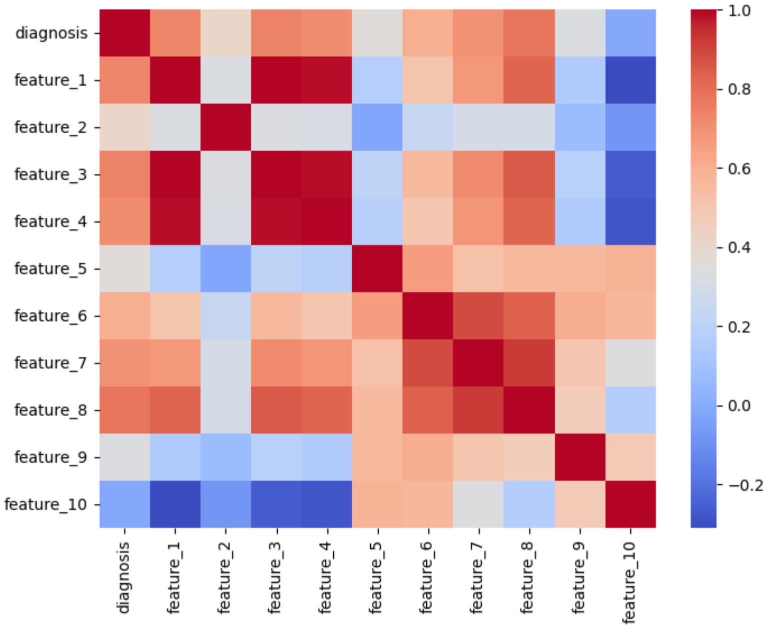


Figure 2: Correlation heatmap of the diagnosis label with the first ten features.

**Inference:** The heatmap shows very strong positive correlation among `feature_1` (radius), `feature_3` (perimeter) and `feature_4` (area), which is expected since these three quantities are geometrically dependent on each other. `feature_2` (texture) and `feature_10` (fractal dimension) are comparatively weakly correlated with the diagnosis and with the size-related features. This redundancy among size-related features suggests that tree-based models, which naturally perform implicit feature selection at each split, are well suited to this dataset since they can ignore redundant correlated features without requiring explicit dimensionality reduction.

## 5 Data Preprocessing

- **Missing value handling:** Not required, as the WDBC dataset has no missing entries.
- **Encoding:** The categorical `diagnosis` column was label-encoded using `LabelEncoder`, mapping Benign to 0 and Malignant to 1. The non-predictive `id` column was dropped.
- **Feature scaling:** Not applied, since both Decision Tree and Random Forest are split-based, scale-invariant algorithms that do not require standardized or normalized input features.
- **Train-test split:** The data was split into 80% training (455 samples) and 20% testing (114 samples) using `train_test_split` with `random_state=42`.
- **Feature engineering:** No additional feature engineering was performed; all 30 pre-computed nucleus-measurement features were used directly.

# 6 Machine Learning Algorithm

## Decision Tree

**Working principle:** A Decision Tree recursively splits the feature space into regions that are increasingly pure with respect to the class label, choosing at each node the feature and threshold that best separates the classes according to an impurity criterion.

**Mathematical formulation:** For a node with class proportions $p_1, \ldots, p_k$, the Gini impurity and entropy criteria used for splitting are

$$\text{Gini} = 1 - \sum_{i=1}^{k} p_i^2, \qquad \text{Entropy} = -\sum_{i=1}^{k} p_i \log_2 p_i \tag{1}$$

At each node, the split that maximizes the impurity reduction (information gain) is selected.

**Advantages:** Easy to interpret and visualize, requires no feature scaling, and naturally handles non-linear decision boundaries.

**Limitations:** Prone to overfitting when grown too deep, and individual trees can be unstable (sensitive to small changes in the training data).

## Random Forest

**Working principle:** A Random Forest is an ensemble of decision trees, each trained on a bootstrap sample of the data and considering only a random subset of features at each split. Predictions are aggregated by majority vote (classification).

**Mathematical formulation:** For $T$ trees $h_1, \ldots, h_T$, the ensemble prediction is

$$\hat{y} = \text{mode}\{h_1(\mathbf{x}), h_2(\mathbf{x}), \ldots, h_T(\mathbf{x})\} \tag{2}$$

Bootstrap aggregation (bagging) combined with random feature selection at each split reduces the correlation between individual trees, which lowers the variance of the ensemble compared to a single decision tree.

**Advantages:** More robust to overfitting than a single tree, generally higher accuracy, and provides a built-in measure of feature importance.

**Limitations:** Less interpretable than a single decision tree, higher memory and computation cost, and prediction time grows with the number of trees.

# 7 Experimental Setup

| | |
|---|---|
| Python Version | 3.11 |
| Scikit-Learn Version | 1.4 |
| IDE | Jupyter Notebook |
| Operating System | Windows / Linux (as used at runtime) |
| Hardware | Standard laptop / cloud CPU runtime, no GPU required |

# 8 Hyperparameter Selection

| Parameter | Value |
|---|---|
| Decision Tree – search grid | criterion: [gini, entropy]; max_depth: [3, 5, 7, None]; min_samples_split: [2, 5]; min_samples_leaf: [1, 2] |
| Decision Tree – best parameters | criterion = entropy, max_depth = 3, min_samples_split = 2, min_samples_leaf = 1 |
| Random Forest – search grid | n_estimators: [50, 100]; max_depth: [5, 10, None]; max_features: [sqrt, log2]; bootstrap: [True] |
| Random Forest – best parameters | n_estimators = 100, max_depth = 5, max_features = sqrt, bootstrap = True |
| Cross-validation folds (Grid Search) | 5 |

Both grid searches used `GridSearchCV` with `cv=5` and `scoring='accuracy'`. The best cross-validation accuracy obtained was 0.9429 for the Decision Tree and 0.9604 for the Random Forest.

# 9 Results

| Metric | Decision Tree (tuned) | Random Forest (tuned) |
|---|---|---|
| Accuracy | 0.9649 | 0.9649 |
| Precision | 1.0000 | 0.9756 |
| Recall | 0.9070 | 0.9302 |
| F1-score | 0.9512 | 0.9524 |
| ROC-AUC | 0.972 | 0.996 |

# 10 Result Visualization

## 5-Fold Cross-Validation Accuracy

| Model | Fold 1 | Fold 2 | Fold 3 | Fold 4 | Fold 5 | Average |
|---|---|---|---|---|---|---|
| Decision Tree | 0.9035 | 0.9123 | 0.9561 | 0.9386 | 0.9558 | 0.9333 |
| Random Forest | 0.9211 | 0.9386 | 0.9825 | 0.9737 | 0.9823 | 0.9596 |

**Inference:** The Random Forest outperforms the Decision Tree on every single fold, with an average cross-validation accuracy of 0.9596 compared to 0.9333 for the Decision Tree. The gap is largest in folds 3 through 5, suggesting the ensemble is more consistently robust to different train/validation splits. The lower fold-to-fold variance for the Random Forest also indicates that bagging and feature randomization are effectively reducing the variance that a single decision tree exhibits.

## Decision Tree Confusion Matrix

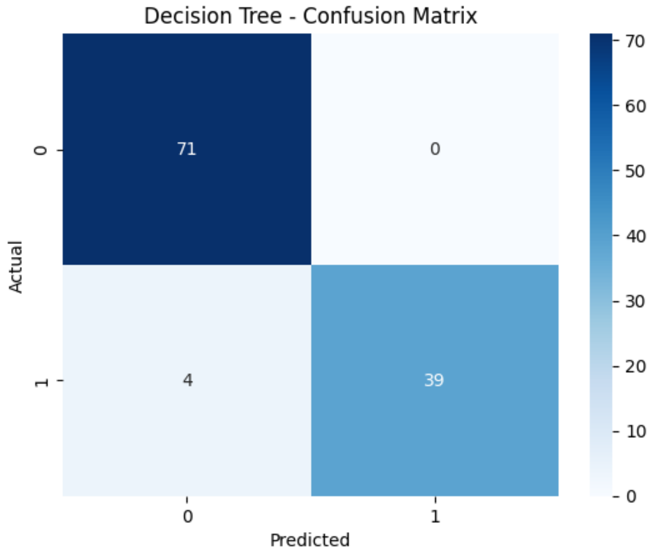


Figure 3: Confusion matrix for the tuned Decision Tree on the test set.

**Inference:** The Decision Tree correctly classifies all 71 benign cases (precision = 1.0) but misses 4 malignant cases, classifying them as benign (recall = 0.907). In a medical diagnosis context, these false negatives are the more costly type of error, since a missed malignant tumour delays treatment. This shows that while the tree is very conservative about labelling a case malignant, it does so at the cost of missing some true malignant cases.

## Random Forest Confusion Matrix

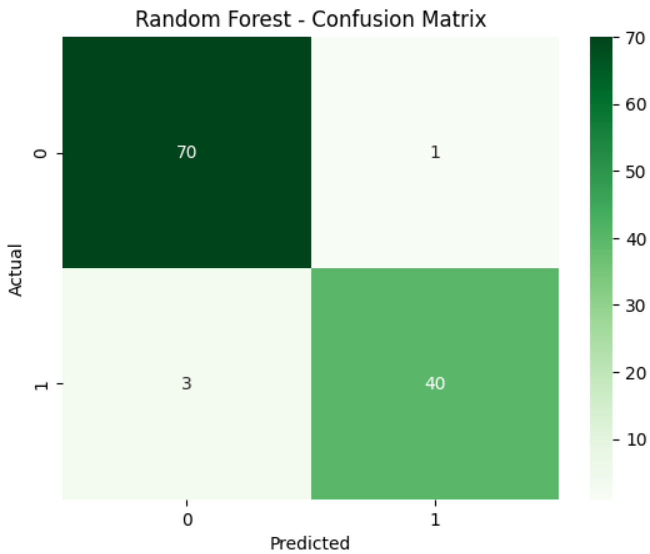


Figure 4: Confusion matrix for the tuned Random Forest on the test set.

**Inference:** The Random Forest misses only 3 malignant cases (recall = 0.930) while misclassifying 1 benign case as malignant (precision = 0.976). Compared to the Decision Tree, the Random Forest trades a small amount of precision for a meaningful gain in recall, catching one more true malignant case out of the four the Decision Tree missed. For a cancer-screening task, this improvement in recall is generally more valuable than the small drop in precision.

## ROC Curve

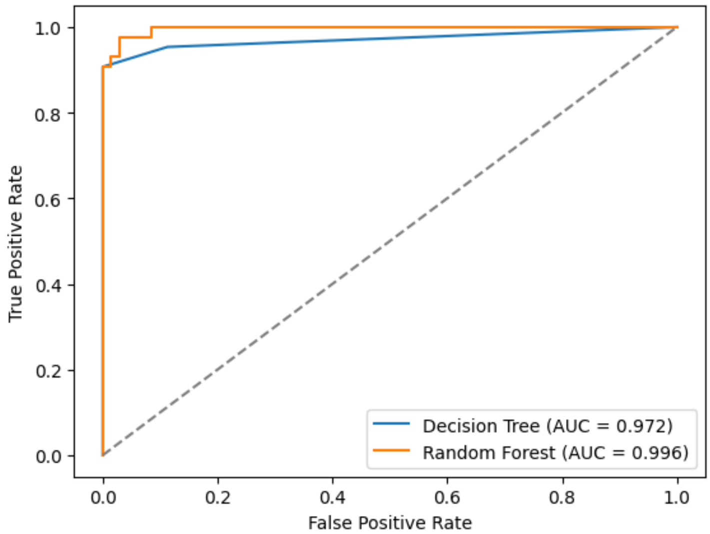


Figure 5: ROC curves for the tuned Decision Tree and Random Forest on the test set.

**Inference:** The Random Forest achieves a near-perfect AUC of 0.996, compared to 0.972 for the Decision Tree, confirming that the ensemble separates the two classes more effectively across all classification thresholds, not just at the default threshold used for the confusion matrices. The Decision Tree's ROC curve rises quickly at very low false-positive rates but plateaus earlier than the Random Forest's curve, reflecting the single tree's coarser decision boundary. This gap in AUC reinforces the cross-validation result that the Random Forest generalizes more reliably than the single Decision Tree on this dataset.

### Effect of Tree Depth on Overfitting

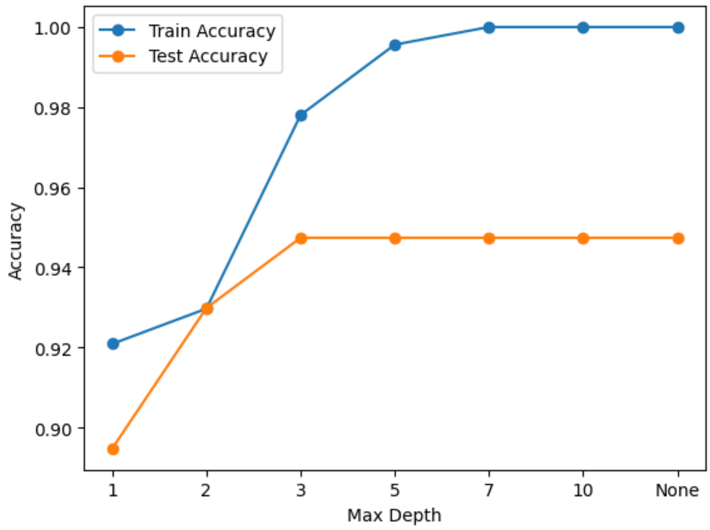


Figure 6: Training versus test accuracy of a Decision Tree as `max_depth` increases.

**Inference:** As `max_depth` increases from 1 to 3, both training and test accuracy rise sharply. Beyond a depth of about 3, training accuracy continues climbing until it reaches 1.0 (perfect fit on training data) at depth 5 and beyond, while test accuracy plateaus around 0.947 and does not improve further. This growing gap between training and test accuracy is a clear signature of overfitting: an unconstrained Decision Tree memorizes the training data without generalizing better to unseen data. This directly explains why the grid search selected a shallow tree (`max_depth = 3`) as the best-performing configuration, since it balances fit and generalization.

# 11 Discussion

Both classifiers achieve identical test accuracy (0.9649), but a closer look at the other metrics shows meaningful differences. The Decision Tree achieves perfect precision (1.0) but lower recall (0.907), meaning it never raises a false alarm but misses more true malignant cases. The Random Forest achieves a more balanced precision-recall trade-off (0.976 / 0.930) and a substantially higher ROC-AUC (0.996 vs. 0.972), as well as higher and more stable cross-validation accuracy (0.960 vs. 0.933 average, with lower variance across folds). The overfitting analysis further explains why a single decision tree is inherently limited on this dataset: an unconstrained tree overfits almost immediately, and even the tuned, depth-limited tree cannot match the generalization ability of an ensemble of many such trees. Overall, the Random Forest is the more reliable classifier for this diagnostic task, primarily because of its higher recall and AUC, both of which matter more than a marginal precision loss in a cancer screening setting. A limitation of this experiment is that only accuracy was used as the grid-search scoring metric; using recall or F1-score as the scoring criterion could yield different, potentially better-suited hyperparameters for a screening application where false negatives are especially costly.

## 12 Conclusion

This experiment implemented and compared Decision Tree and Random Forest classifiers for breast cancer diagnosis using the Wisconsin Diagnostic Breast Cancer dataset. After label encoding and an 80/20 train-test split, both models were tuned using 5-fold grid search and validated using 5-fold cross-validation. While both models reached the same test accuracy (96.49%), the Random Forest demonstrated superior recall, ROC-AUC (0.996), and cross-validation stability compared to the single Decision Tree, and the overfitting analysis confirmed that unconstrained decision trees overfit the training data quickly. The Random Forest is therefore the recommended model for this diagnostic task, with the Decision Tree retained as a useful, highly interpretable baseline for understanding which individual features drive the diagnosis.

## 13 References

1. B. Sugith, "Experiment_5.ipynb," GitHub Repository, 2025. [Online]. Available: https://github.com/sugith-2025/MachineLearning/blob/main/Experiment_5.ipynb